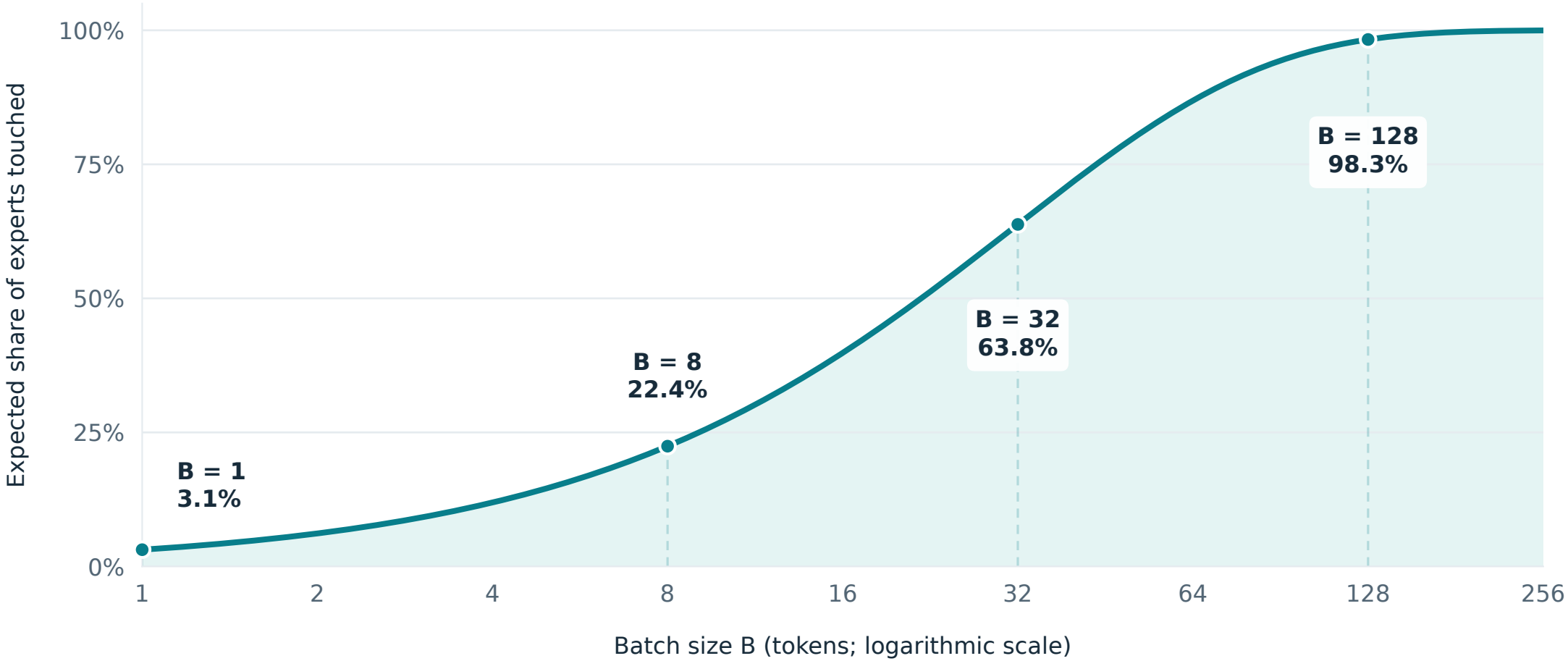
# A larger batch touches more experts

Idealized MoE routing | 256 equal-size experts | 8 experts per token




$$\mathbb{E}[U_B]/N = 1 - (1 - k/N)^B$$

Uniform routing • Independent tokens • Equal-size experts

Analytical expectation; illustrative model, not a measured benchmark.